👩‍🔬 **Tester:** Paula Granado

🎯 **API Test Objective:**
Validate the integrity and functionality of the Stripe API's core payment flow. Focus on ensuring correct creation and confirmation of Payment Intents, validating HTTP status codes, and ensuring robust error handling for negative scenarios and authentication failures.

🔒 **API Test Prerequisites**
- Access to a Stripe Dashboard (Test Mode).
- A valid Secret API Key (sk_test_...) configured in Postman environments.
- Stable internet connection for real-time API communication.

🧪 **API Test Endpoints (v1)**
- POST /v1/payment_intents: Create a new payment intent.
- POST /v1/payment_intents/{id}/confirm: Confirm a payment intent.

👥 **Test Scenarios & Profiles**
- Happy Path: Successful creation and confirmation using valid test cards (e.g., 4242...)
- Negative Scenarios:
    - Declined cards (simulating card errors)
    - Missing mandatory parameters (e.g., amount, currency)
    - Idempotency/State errors (confirming an already processed intent)
Security/Auth: Unauthorized requests (invalid/missing Bearer Token)

🛠️ **Tools Used:**
- Postman: Collection organization and automated test scripts (JavaScript)
- Stripe API Documentation: Reference for expected behaviors and error codes

🔐 **Customer Functionality: Customer Management**
Validate the complete lifecycle of a Customer resource (CRUD), ensuring data integrity, correct mapping of metadata, and proper API response behavior for each operation.

Prerequisites:
- The Stripe API must be responsive (Test Mode).
- Postman configured with Environment Variables for the base_url and secret_key.
- Authentication via Bearer Token must be active.

Test Endpoints:
POST /v1/customers: Creates a new customer.
GET /v1/customers/{id}: Retrieves details using a specific Customer ID.
POST /v1/customers/{id}: Updates existing customer attributes (e.g., email, description).
DELETE /v1/customers/{id}: Permanently removes the customer or marks it as deleted.

Category	ID	Test Name	Objective	Precondition	Steps	BDD (Gherkin Syntax)	Expected Result	Actual Result (Occurrence)	Status	Priority	Severity
Happy Path - Customer	TC01	Create Customer (Valid)	Validate that a customer can be created with valid data (name and email).	A valid Secret API Key is available and configured in the Postman Authorization tab (using Basic Auth).	1. Send a POST request to https://api.stripe.com/v1/customers. 2. Add a Body using the x-www-form-urlencoded format with the following data: name: Milan Rosen and email: Milano12@example.com. 3. Send the request.	Given valid customer data When the request is sent Then the API returns a 200 status code and a valid customer object with an ID	The response status code should be 200 OK. The response body should be a JSON object containing a unique id, and the name and email matching the data sent in the request.	As expected.	Passed	High	Critical
Negative Testing - Customer	TC02	Create Customer (Invalid Email)	To validate that the API returns a proper error response when the email provided is not in a valid format.	A valid Secret API Key is available and configured in the Postman Authorization tab (using Basic Auth).	1. Send a POST request to https://api.stripe.com/v1/customers. 2. Add a Body using the x-www-form-urlencoded format with the following data: name: Declan Luke and email: invalid_email. 3. Send the request.	Given invalid data with an improperly formatted email When a POST request is sent to create a customer Then the API returns a 400 status code and a JSON error message	The response status code should be 400 Bad Request. The response body should be a JSON object with an error key containing a code such as "email_invalid" and a descriptive message indicating the email is not valid.	As expected.	Passed	Medium	Major
Happy Path - Customer	TC03	Retrieve Customer	To validate that the API can successfully retrieve the details of an existing customer using their unique ID.	A valid Secret API Key is available. A customer has been successfully created (e.g., from STRP01), and their customer_id is available.	1. Send a GET request to the endpoint: https://api.stripe.com/v1/customers/{customer_id}. 2. Send the request.	Given an existing customer with a valid ID When a GET request is sent to retrieve their details Then the API returns a 200 status code and the correct customer information	The response status code should be 200 OK. The response body should be a JSON object containing the correct customer_id, and the name and email properties should match the data from the creation test (STRP01).	As expected.	Passed	High	Critical
Happy Path - Customer	TC04	Delete Customer	To validate that the API can successfully delete an existing customer resource using their unique ID.	A valid Secret API Key is available. A customer has been successfully created (e.g., from STRP01), and their customer_id is available.	1. Send a DELETE request to the endpoint: https://api.stripe.com/v1/customers/{customer_id}. 2. Send the request.	Given an existing customer with a valid ID When a DELETE request is sent Then the API returns a 200 status code with a confirmation that the customer has been deleted	The response status code should be 200 OK. The response body should be a JSON object containing the correct id of the customer and the property deleted with a value of true.	As expected.	Passed	Medium	Major
Customer	TC05	Update Customer	To validate that an existing customer's data can be successfully updated.	A customer has been successfully created (e.g., from STRP01), and their customer_id is available.	1. Send a POST request to https://api.stripe.com/v1/customers/{customer_id}. 2. Add the Body with the data to update (e.g., email: new.email@example.com). 3. Send the request. 4. Send a GET request to the same endpoint to verify the update.	Given an existing customer When a POST request is sent to update their data Then the API returns a 200 status code and the customer's data is updated	The response to the POST request should be 200 OK. The response to the subsequent GET request should show the updated email property.	As expected.	Passed	High	Critical
Customer	TC06	Retrieve Customer (Non-Existent)	To validate that the API returns a correct error when attempting to retrieve a customer that does not exist.	A valid Secret API Key is available. A customer with ID cus_invalid does not exist.	1. Send a GET request to https://api.stripe.com/v1/customers/cus_invalid. 2. Send the request.	Given a non-existent customer ID When a GET request is sent to retrieve it Then the API returns a 404 status code and an error message	The response status code should be 404 Not Found. The response body should be a JSON object with a clear error message indicating that the resource was not found.	As expected.	Passed	Medium	Major

Payment Functionality: Payment Flow
Validate the end-to-end payment lifecycle, ensuring that Payment Intents are correctly initialized, transitioned through states, and successfully confirmed. The tests cover success paths (Happy Path) and common failure scenarios like declined payments or missing parameters.

Prerequisites:
-The Stripe API must be responsive in Test Mode.
- Postman configured with a valid Secret Key via Bearer Token.
- Use of Stripe Test Cards (e.g., 4242...) for simulation.

Test Endpoints:
POST /v1/payment_intents: Creates a new intent with specified amount and currency.
POST /v1/payment_intents/{id}/confirm: Finalizes the payment using a provided payment_method.

Category	ID	Test Name	Objective	Precondition	Steps	BDD (Gherkin Syntax)	Expected Result	Actual Result (Occurrence)	Status	Priority	Severity
Happy Path - Payment	TC07	Create Payment Intent	To validate that a Payment Intent can be successfully created with valid data, which is the first step in a payment transaction.	A valid Secret API Key is available and configured in the Postman Authorization tab (using Basic Auth).	1. Send a POST request to https://api.stripe.com/v1/payment_intents. 2. In the Body, use x-www-form-urlencoded with the following key-value pairs: • amount: 2000 (for €20.00) • currency: eur • payment_method_types[]: card 3. Send the request.	Given valid payment data When a POST request is sent to create a Payment Intent Then the API returns a 200 status code and a Payment Intent object with a requires_payment_method status	The response status code should be 200 OK. The response body should be a JSON object containing a unique id, and a status property with the value requires_payment_method. The amount and currency should also match the request.	As expected.	Passed	High	Critical
Happy Path - Payment	TC08	Confirm Payment Intent	To validate that a Payment Intent can be successfully confirmed using a valid test card, completing the transaction flow.	A valid Secret API Key is available. A Payment Intent has been successfully created (e.g., from STRP05), and its payment_intent_id is available.	1. Send a POST request to https://api.stripe.com/v1/payment_intents/{payment_intent_id}/confirm. 2. In the Body, use x-www-form-urlencoded with the key-value pair: payment_method: pm_card_visa. 3. Send the request.	Given a valid Payment Intent and a valid test card When a POST request is sent to confirm the payment Then the API returns a 200 status code and the payment status is succeeded	The response status code should be 200 OK. The response body should be a JSON object containing a unique id and a status property with the value succeeded. The amount and currency should also match the original Payment Intent.	The API returned 200 OK, but the status in the response body was requires_confirmation. In the Stripe dashboard, the payment appears as incomplete. This indicates that the payment method has been associated, but the transaction has not yet been captured.	Passed	High	Critical
Negative Testing - Payment	TC09	Confirm Payment (Declined Card)	To validate that a Payment Intent correctly fails and returns an appropriate error when a declined test card is used.	A valid Secret API Key is available. A Payment Intent has been successfully created (e.g., from STRP05), and its payment_intent_id is available.	1. Send a POST request to https://api.stripe.com/v1/payment_intents/{payment_intent_id}/confirm. 2. In the Body, use x-www-form-urlencoded with the key-value pair: payment_method: pm_card_chargeDeclined. 3. Send the request.	Given a valid Payment Intent and a declined test card When a POST request is sent to confirm the payment Then the API returns a 402 status code and an error message with card_declined	The response status code should be 402 Payment Required. The response body should be a JSON object with an error key containing a code: "card_declined" and a descriptive message for the user.	The API returned a 402 Payment Required status code. The response body contained an error object with the code card_declined. This proves the API correctly handled the declined payment scenario and provided the expected error information.	Passed	Medium	Major
Negative Testing - Payment	TC10	Create Payment Intent (Missing Amount)	To validate that the API returns a proper error when a required field (amount) is missing from the request body.	A valid Secret API Key is available.	1. Send a POST request to https://api.stripe.com/v1/payment_intents. 2. In the Body, use x-www-form-urlencoded with only the following data: currency: usd and payment_method_types[]: card. 3. Send the request.	Given a request with a missing required field When a POST request is sent to create a Payment Intent Then the API returns a 400 status code and a JSON error message	The response status code should be 400 Bad Request. The response body should contain a specific error indicating that the amount field is required.	As expected.	Passed	Medium	Major
Negative Testing - Payment	TC11	Create Payment Intent (Invalid Amount)	To validate that the API rejects a transaction when the amount is invalid (e.g., negative).	A valid Secret API Key is available.	1. Send a POST request to https://api.stripe.com/v1/payment_intents. 2. In the Body, use x-www-form-urlencoded with the following data: amount: -100, currency: usd and payment_method_types[]: card. 3. Send the request.	Given a request with an invalid amount When a POST request is sent to create a Payment Intent Then the API returns a 400 status code and an error message	The response status code should be 400 Bad Request. The response body should contain a JSON object with a specific error message stating that the amount must be a positive integer.	As expected.	Passed	Medium	Major
Negative Testing - Payment	TC12	Confirm Payment (Already Confirmed)	To validate that the API prevents a Payment Intent from being confirmed more than once.	A valid Secret API Key is available. A Payment Intent has been successfully created and confirmed (e.g., from STRP06), and its payment_intent_id is available.	1. Send a POST request to https://api.stripe.com/v1/payment_intents/{payment_intent_id}/confirm. 2. In the Body, use x-www-form-urlencoded with the key-value pair: payment_method: pm_card_visa. 3. Send the request.	Given a Payment Intent that has already been confirmed When a POST request is sent to confirm it again Then the API returns a 400 status code with an error message	The response status code should be 400 Bad Request. The response body should contain a JSON object with an error message indicating that the Payment Intent cannot be confirmed again because it has already been processed.	As expected.	Passed	High	Critical